

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	Total Marks:	100
CO2 Statement:	<i>Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)</i>		
Student Name:	Avinash	Reg. No.:	192411044
Section / Batch:	CSE	Date:	27-07-2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given scenario and identify the computational problem involved.
- Apply appropriate Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems wherever applicable.
- Clearly justify the chosen solution approach with proper reasoning and analytical interpretation.
- Demonstrate decision-making skills by comparing possible solutions and selecting the most suitable approach.
- Use diagrams, stepwise illustrations, or algorithmic representations wherever necessary.
- Maintain clarity, logical organization, and proper technical presentation throughout the answers.

Question Type	Focus Area	Questions
Scenario Based Assessment	Analyze real-world scenarios and apply Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems	Q1

SECTION A – SCENARIO BASED ASSESSMENT

Q1. A logistics company computes distances between delivery hubs to identify the closest ones using brute-force computation.

- Explain how all pairs are evaluated.
- Derive the time complexity.
- Discuss scalability concerns.

Code of Conduct

I Avinash K (192411044) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Avinash

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25%	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario
Application of Concepts	30%	Applies appropriate concepts effectively to solve complex real-world problems	Applies relevant concepts correctly with minor errors	Applies basic concepts but with limited accuracy	Fails to apply appropriate concepts
Analytical & Decision-Making Skills	20%	Demonstrates strong analysis and selects optimal solutions with justification	Good analysis with reasonable solutions	Limited analysis; solutions lack justification	Poor or no analysis; incorrect decisions
Solution Design & Approach	15%	Well-structured, innovative, and feasible solution approach	Logical and structured solution with minor gaps	Basic solution with limited structure or feasibility	Unclear or incorrect solution approach
Clarity, Justification & Presentation	10%	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25%				
Application of Concepts	30%				
Analytical & Decision-Making Skills	20%				
Solution Design & Approach	15%				
Clarity, Justification & Presentation	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

```

#include <stdio.h>
#include <math.h>

int main() {
    int n, i, j;
    float x[100], y[100];
    float minDist, dist;
    int hub1, hub2;

    printf("Enter number of delivery hubs: ");
    scanf("%d", &n);

    printf("Enter coordinates (x y) of each hub:\n");
    for(i = 0; i < n; i++) {
        scanf("%f %f", &x[i], &y[i]);
    }

    minDist = 1e9;

    printf("\nDistances between all pairs:\n");

    for(i = 0; i < n - 1; i++) {
        for(j = i + 1; j < n; j++) {

            dist = sqrt((x[j] - x[i]) * (x[j] - x[i]) +
                (y[j] - y[i]) * (y[j] - y[i]));

            printf("Hub %d - Hub %d = %.2f\n", i + 1, j + 1, dist);

            if(dist < minDist) {
                minDist = dist;
                hub1 = i;
                hub2 = j;
            }
        }
    }

    printf("\nClosest Hubs: Hub %d and Hub %d\n", hub1 + 1, hub2 + 1);
    printf("Minimum Distance = %.2f\n", minDist);

    return 0;
}

```

Enter coordinates (x y) of each hub:

1 1

4 5

7 2

3 1

Distances between all pairs:

Hub 1 - Hub 2 = 5.00

Hub 1 - Hub 3 = 6.08

Hub 1 - Hub 4 = 2.00

Hub 2 - Hub 3 = 4.24

Hub 2 - Hub 4 = 4.12

Hub 3 - Hub 4 = 4.12

Closest Hubs: Hub 1 and Hub 4

Minimum Distance = 2.00

Process exited after 33.51 seconds with return value 0

Press any key to continue . . . |